Exam 3 review

ECON-4210

Economics of Climate Change

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

- Ghost Emissions round 8: where does the money go?
- Exam 3: format and coverage
- Problem set 3 questions?
- Practice problems

Ghost Emissions Rd 8: where does the money go?

- In round 8, each borough decided what to do with its budget (1/5 of income every round):
  - subsidize containment
  - pay for adaptation
  - fund trap R&D
  - or hold in reserve

- In the game, businesses pay the whole tax. Their output price is fixed, so they cannot pass any of it on.
  - In real markets, much of a fuel tax shows up in the prices consumers pay.

- Which boroughs recycled the budget into containment subsidies? What happened to containment and to businesses' costs?

What did mayors do in round 8?

Borough	Price on ghosts this round	What the mayor did with the budget
Coalbrook	$300 tax	Containment subsidy and R&D
Marsh End	$100 tax	All of it on R&D
The Stacks	$90 tax	Slime defense, R&D, some subsidy
Fenwick Island	22% containment rule	Split evenly: subsidy, defense, R&D
Northgate	$20 tax	Mostly slime defense
Old Exchange	$30 tax	Mostly defense, some subsidy and R&D
Midtown Common	$200 tax	A little of each, half held in reserve
Lumen Heights	$100 tax	Some subsidy, most held in reserve
Harborline	$100 tax	Nothing. No decision submitted

The budget is 1/5 of borough income each round. Tax revenue goes straight to borough welfare.

Georgia Institute of Technology

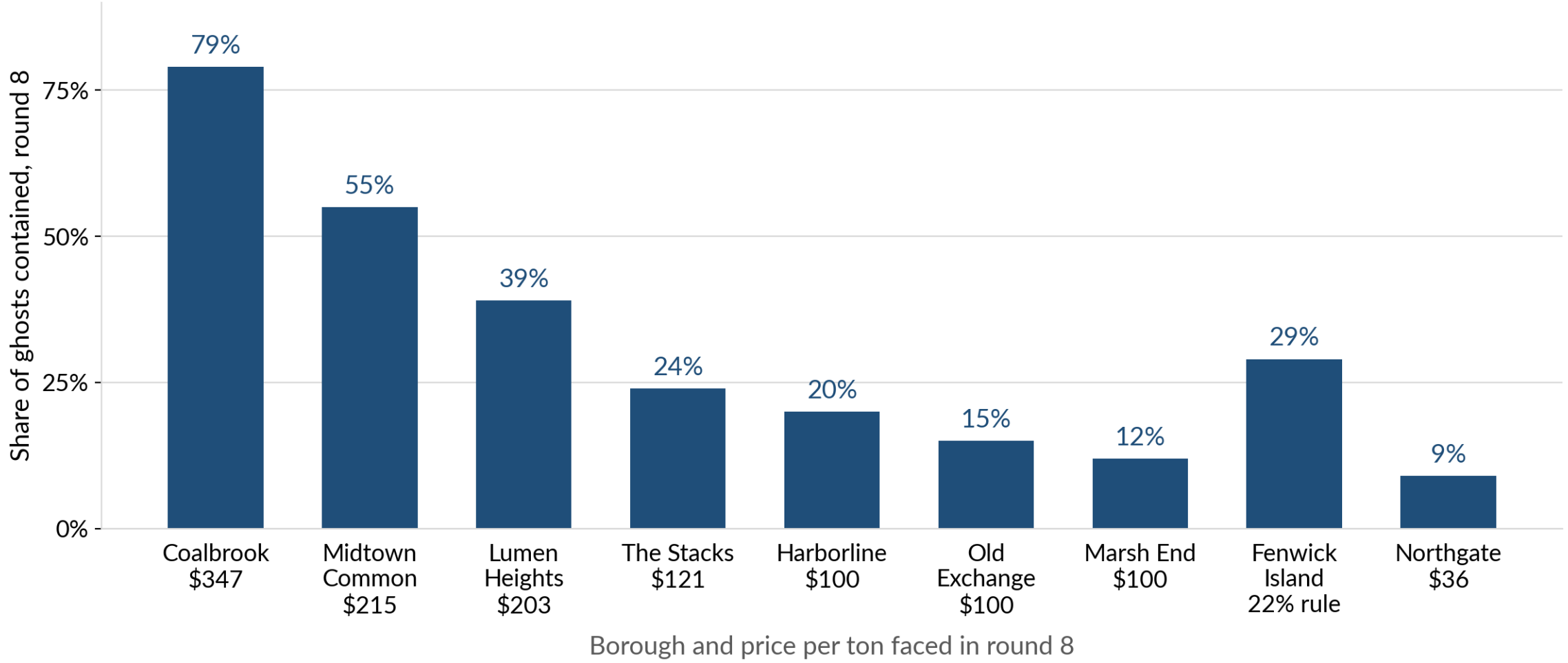
What did businesses do in response?

Borough	Price per ton, round 7 → 8	Containment, round 7 → 8	Ghosts released, round 7 → 8
Coalbrook	$50 → $347	33% → 79%	64 → 18
Midtown Common	$110 → $215	37% → 55%	55 → 37
Lumen Heights	25% rule → $203 (tax + subsidy)	39% → 39%	36 → 36
The Stacks	$170 → $121	39% → 24%	70 → 86
Harborline	$178 permit → $100	36% → 20%	43 → 64
Marsh End	$40 → $100	11% → 12%	70 → 66
Old Exchange	none → $100 (tax + subsidy)	19% → 15%	59 → 61
Northgate	$90 → $36	29% → 9%	72 → 99
Fenwick Island	$130 → 22% rule (+ $62 subsidy)	36% → 29%	28 → 30

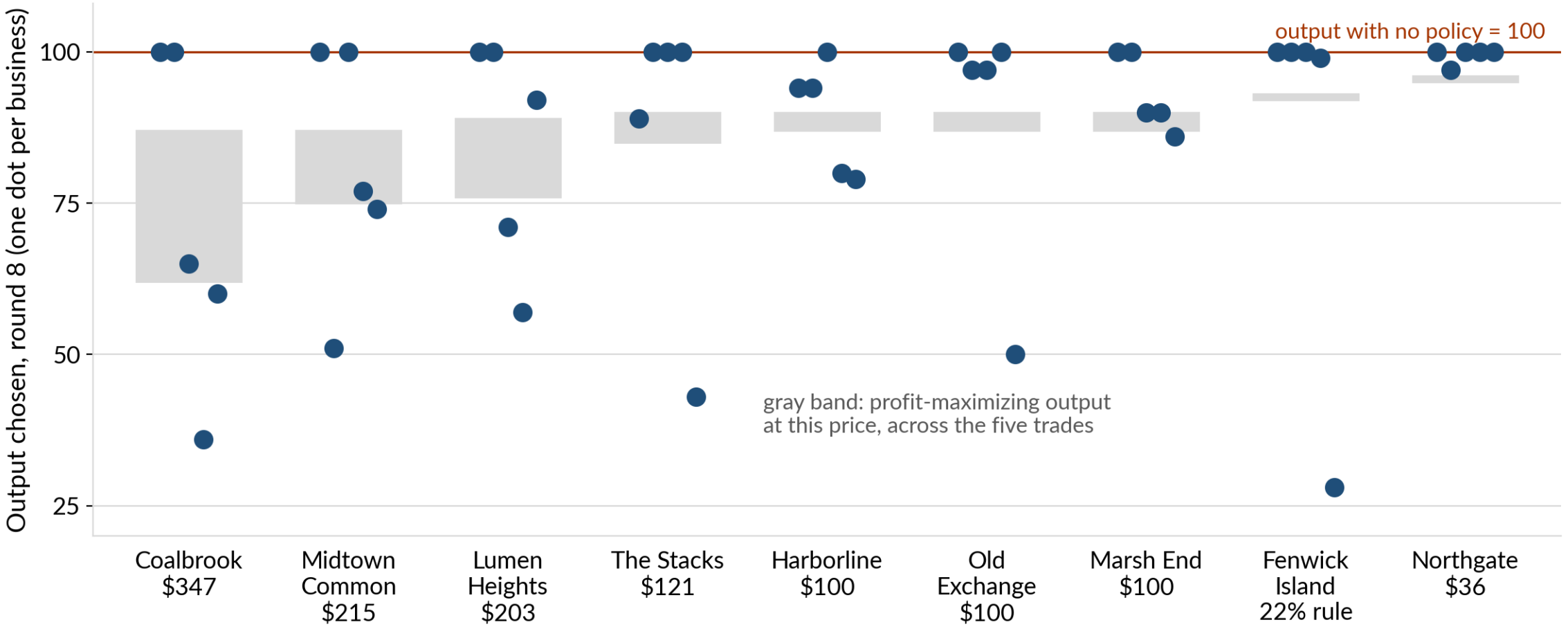
For big increases in emissions price, businesses contained more.
For emissions price decreases, businesses contained less.

Georgia Institute of Technology

Containment in round 8 by price



Output in round 8, by the price faced



A price on ghosts works on two margins: contain more per unit, and make fewer units. Observations above the gray band left money on the table.

Exam 3

Monday, November 16, in class

- 20 multiple choice questions, 1 point each.
- 2 short answer questions, 10 points each.
  - Show your work. Partial credit for a correct method with an arithmetic error.
- One-page cheat sheet. No other references, books, or technology.
- You may use a calculator.

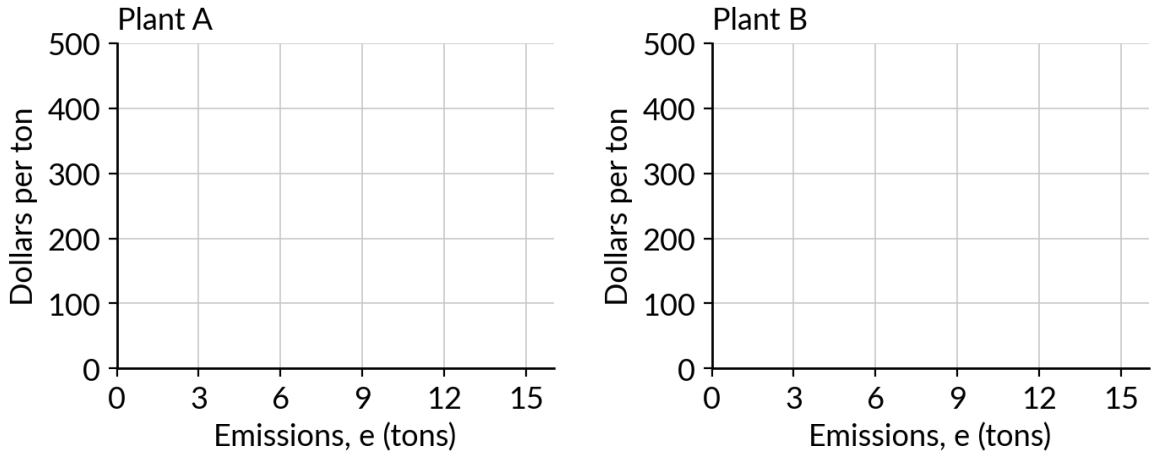
What does the exam cover?

Classes 15–20 and Problem Set 3

- 15 — How would an economist solve climate change? Standards and taxes
- 16 — Cap-and-trade
- 17 — Prices vs. quantities, innovation, and carbon pricing in practice
- 18 — In-class policy game
- 19 — The permits game: debrief
- 20 — Who pays for a carbon tax?
  - Readings we've covered in class are fair game.

Practice problem 1

- Two power plants each emit 15 tons of CO_2 without regulation.
- Each plant's MAC depends on its emissions, e:
  - **$MAC_A = 240 - 16e_A$**
  - **$MAC_B = 480 - 32e_B$**
  - EPA wants to cap total emissions = 12 tons.



a) Uniform standard: each plant emits 6 tons. Last-ton MAC and TAC for each plant?

b) What tax rate produces 12 tons? Emissions and TAC for each plant?

c) The EPA gives all 12 permits to plant B. Who buys, how many, at what price? Net cost for each plant?

d) Which policy costs the least? Why?

Practice problem 1

TAC is the area under MAC between a plant's emissions and 15 tons.

Policy	e_A	e_B	TAC_A	TAC_B	Total TAC
(a) Standard, 6 tons each	6	6			
(b) Tax					
(c) All permits to B, after trading					

Practice problem 2

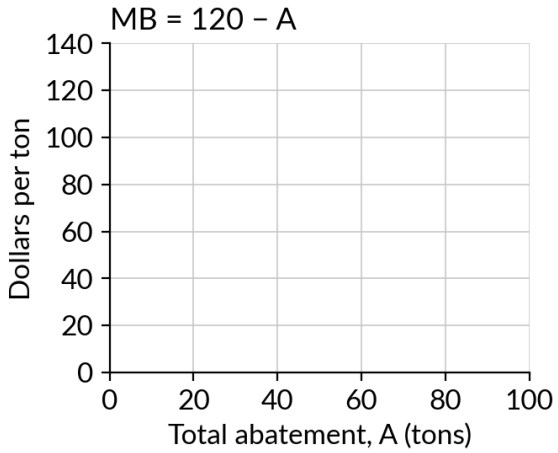
- A regulator expects MAC = A, where A is total tons abated.
- It sets a tax or a cap where expected MAC intersects MB.
- Abatement costs turn out 50% higher than expected:
  - **MB = 120 – A**
  - **True MAC = 1.5A**

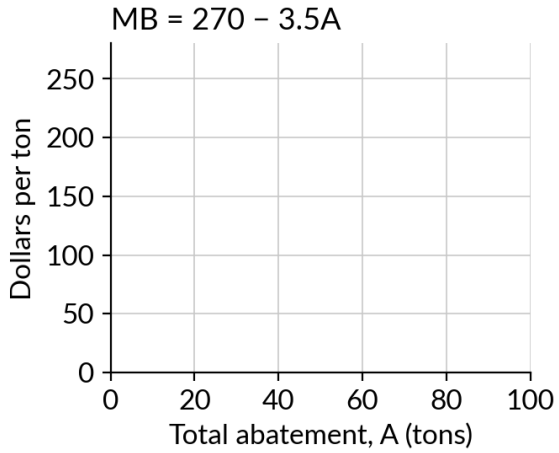
a) What tax and what cap does the regulator set?

b) How much do firms abate under each policy? What is the efficient level?

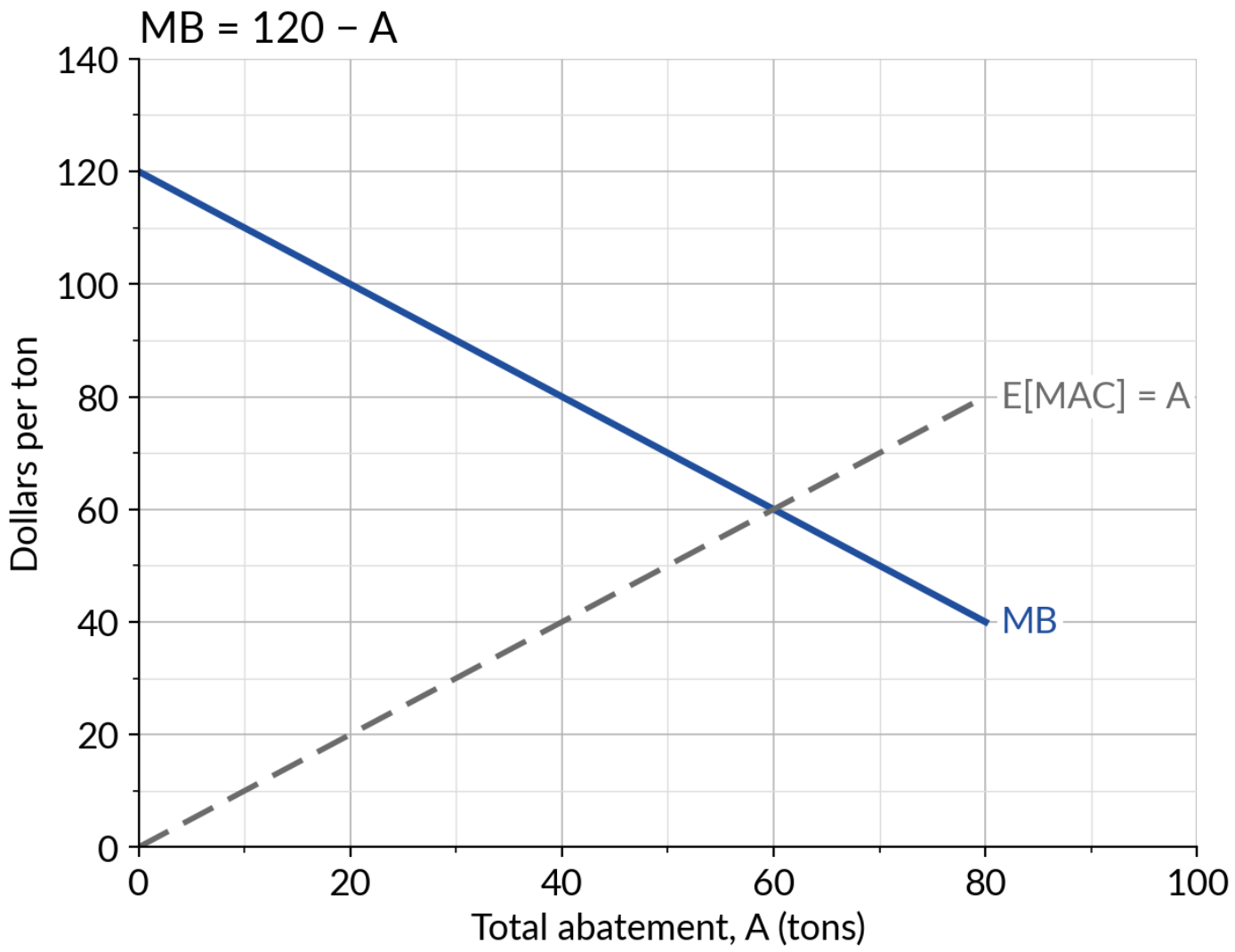
c) What is the deadweight loss under each policy? Which is smaller?

d) Repeat with MB = 270 − 3.5A. Which case looks more like climate change?





Practice problem 2

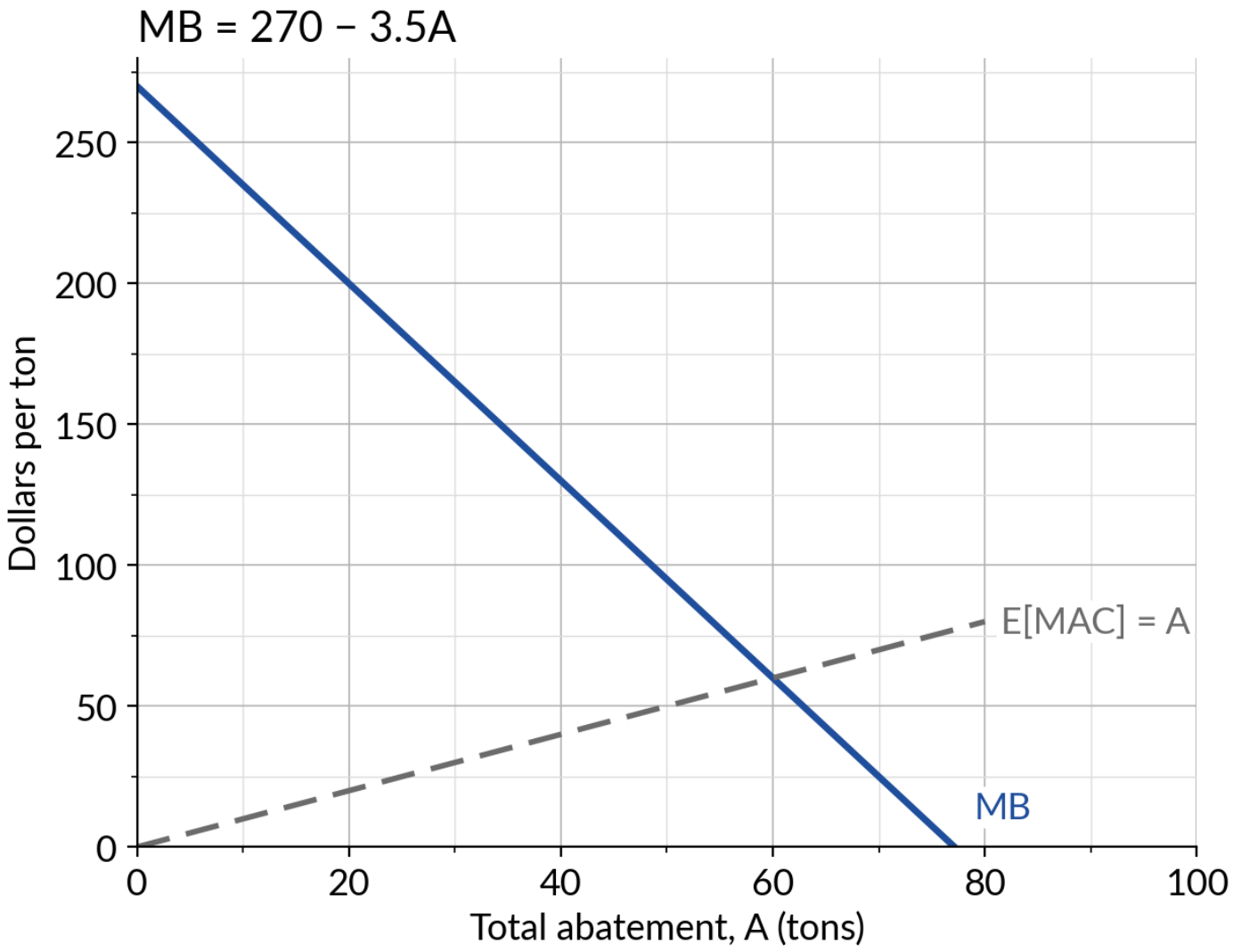


Expected MAC = A

True MAC = 1.5A

Georgia Institute of Technology

Practice problem 2



Expected MAC = A

True MAC = 1.5A

Practice problem 3

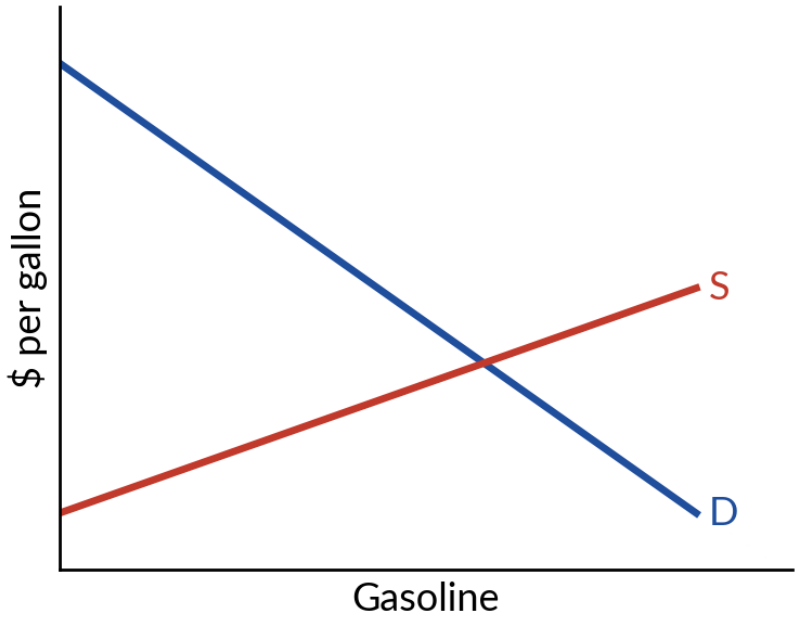
- Drivers buy 100 billion gallons of gasoline a year at $3.00.
- The government levies a carbon tax:
  - **$\varepsilon_D = -0.6$, $\varepsilon_S = 0.4$**
  - **Tax = $1 per gallon**

a) What share of the tax do consumers bear? What do consumers pay and producers keep?

b) New quantity? Tax revenue and the total burden on each side?

c) Revenue is returned as an equal dividend to 115 million households. Dividend per household?

d) Household A buys 2,500 gallons a year, household B buys 200. Burden at the pump, net of the dividend? Who pays for the rest?



Georgia Institute of Technology

Practice problem 3

Consumers' share = $\varepsilon_S / (\varepsilon_S + |\varepsilon_D|)$

Step	Value
(a) Consumers' share of the tax	
(a) Consumer price, producer price	
(b) Quantity (billion gallons)	
(b) Revenue	
(b) Burden: consumers, producers	
(c) Dividend per household	
(d) Household A: burden, net	
(d) Household B: burden, net	